Naming convention

Filename pattern

Most time-step outputs are named

`<object>_<group>_<interval>.txt`

for example `hru_wb_day.txt` (HRU water balance, daily) or `basin_sd_cha_aa.txt` (basin sum of channel outputs, average annual). Some files have no group (`aquifer_day.txt`, `reservoir_mon.txt`, `ru_yr.txt`), and a few files written once per run have no interval (`hru_cbn_lyr.txt`, `mgt_out.txt`).

Interval	Meaning	print.prt column
`_day`	one row per object per day	`daily`
`_mon`	one row per object per month	`monthly`
`_yr`	one row per object per year	`yearly`
`_aa`	one row per object, average annual over the printed years	`avann`

The channel sediment budget files `ch_sedbud_*` and `chanbud_order_*` use `_d`, `_m`, `_y` and `_a` instead.

When `csvout` is `y` in print.prt, most time-step files also get a comma-separated copy with the same name and the extension `.csv`.

Output folder

Output files are written to the folder named on the output path line at the end of file.cio. SWAT+ creates the folder if it does not exist. When the line is missing, empty or `null`, outputs go to the folder SWAT+ runs in. `simulation.out`, `success.fin` and `erosion.txt` always go to the run folder.

File layout

Time-step files share one layout:

- Line 1: the project name from file.cio and the SWAT+ version banner.
- Line 2: column headers.
- Line 3: units (blank for columns without units).
- Lines 4 and on: data rows.

Rows with one object per row start with `jday`, `mon`, `day`, `yr`, `unit`, `gis_id` and `name`. Basin files use the same columns with `unit` and `gis_id` set to 1. `jday`, `mon`, `day` and `yr` are the date of the last day in the print step.

Print timing

The first blocks of print.prt set when rows are written:

- Daily rows are written from `day_start` of `yrc_start` to `day_end` of `yrc_end`, every `interval` days. A zero `day_start` means day 1, a zero `day_end` means day 366, and zero years mean the first and last simulated year.
- Monthly, yearly and average annual rows start after the first `nyskip` simulated years. Average annual values are averages over the remaining years.

- The average annual print intervals (`aa_int_cnt` and the end years that follow it) are read, but in 62.0.1 they are used only by the region output, which is not written, so they have no effect.

print.prt rows and files

Each row in the objects block of `print.prt` has a name and four `y`/`n` switches. When `use_obj_lbls` is `y`, rows are matched by name, may come in any order, and rows can be left out (they default to `n`). A name that appears twice stops the run with an error, and so does a name SWAT+ does not know. When `use_obj_lbls` is `n`, the rows are read by position and must follow the order of the table below up to `wetland_cs`; the carbon and gwflow rows can only be set in labelled mode.

Each `y` opens the file for that interval. The table shows the file stems; add `_day.txt`, `_mon.txt`, `_yr.txt` or `_aa.txt`.

`print.prt` row	Files	Described in
`basin_wb`	`basin_wb`	Water balance
`basin_nb`	`basin_nb`	Nutrient balance and losses
`basin_ls`	`basin_ls`	Nutrient balance and losses
`basin_pw`	`basin_pw`	Aquifer, reservoir, plant
`basin_aqu`	`basin_aqu`	Aquifer, reservoir, plant
`basin_res`	`basin_res`	Aquifer, reservoir, plant
`basin_cha`	`basin_cha` (legacy channels)	Channel and sediment
`basin_sd_cha`	`basin_sd_cha`, `basin_sd_chamorph`, `basin_sd_chanbud`	Channel and sediment
`basin_psc`	`basin_psc` (recall objects summed)	Recall
`region_wb`, `region_nb`, `region_ls`, `region_pw`, `region_aqu`, `region_res`, `region_sd_cha`, `region_psc`	none in 62.0.1	
`water_allo`	`water_allo`, `wallo_use`, `wallo_treat`	Water allocation
`lsunit_wb`	`lsunit_wb`	Water balance
`lsunit_nb`, `lsunit_ls`, `lsunit_pw`	`lsunit_nb`, `lsunit_ls`, `lsunit_pw`	Nutrient balance and losses, Aquifer, reservoir, plant
`hru_wb`	`hru_wb`; also `hru_path` when constituents are simulated	Water balance
`hru_nb`	`hru_nb`, `hru_ncycle`	Nutrient balance and losses
`hru_ls`	`hru_ls`	Nutrient balance and losses
`hru_pw`	`hru_pw`	Aquifer, reservoir, plant
`hru-lte_wb`, `hru-lte_nb`, `hru-lte_ls`, `hru-lte_pw`	`hru-lte_wb`, `hru-lte_ls`, `hru-lte_pw`	Water balance, Nutrient balance and losses, Aquifer, reservoir, plant
`channel`	`channel` (legacy channels)	Channel and sediment
`channel_sd`	`channel_sd`, `channel_sdmorph`, `sd_chanbud`, `sedbud`, `ch_sedbud`, `chanbud_order`, `channel_sd_subday.txt`	Channel and sediment
`aquifer`	`aquifer`	Aquifer, reservoir, plant
`reservoir`	`reservoir`, `wetland`	Aquifer, reservoir, plant
`recall`	`recall`	Recall
`hyd`	`hydin`, `hydout`, `deposition`	Hydrographs
`ru`	`ru` (routing unit hydrographs)	Hydrographs

print.prt row	Files	Described in
pest	hru_pest, channel_pest, reservoir_pest, aquifer_pest, basin_ls_pest, basin_ch_pest, basin_res_pest, basin_aqu_pest	
basin_salt, hru_salt, ru_salt, aqu_salt, channel_salt, res_salt, wetland_salt	basin_salt, hru_salt, rout_unit_salt, aquifer_salt, channel_salt, reservoir_salt, wetland_salt	Salts
basin_cs, hru_cs, ru_cs, aqu_cs, channel_cs, res_cs	basin_cs, hru_cs, rout_unit_cs, aquifer_cs, channel_cs; res_cs opens both reservoir_cs and wetland_cs	Constituents
wetland_cs	none (wetland constituent files follow res_cs)	
hru_cb, hru_cb_vars, hru_cb_gl, hru_cb_trf, hru_cb_lyr, hru_cb_cpool, hru_cb_npool, hru_cb_plt, hru_cb_flux, hru_cb_drv, hru_cb_dyn, hru_cb_snap, lsu_cb_gl, lsu_cb_trf, lsu_cb_plt	carbon files	Carbon
gwflow_wb, gwflow_flux, gwflow_heat, gwflow_solute, gwflow_obs, gwflow_pump	gwflow files	gwflow

The rows hru-lte_nb and region_* are read but open no files in 62.0.1.

Other switches

The line after csvout in print.prt has four more switches:

Switch	Values	Files
crop_yld	a (average annual), y (yearly), b (both), n	crop_yld_aa.txt, crop_yld_yr.txt, basin_crop_yld_aa.txt, basin_crop_yld_yr.txt; y and b also open yield.out
mgtout	y, n	mgt_out.txt, the log of management operations
hydcon	y, n	hydcon.out, the object connection list (see Hydrographs)
fdcout	y, n	flow_duration_curve.out

cdfout (netCDF output) is read but not used.

Files written every run

File	Content
files_out.out	List of every output file opened in the run, with a short tag.
diagnostics.out	Warnings about missing or unmatched inputs.
checker.out	Soil and HRU parameter summary.
erosion.out	USLE factors per HRU.
area_calc.out	Area checks for routing units and landscape units.
lu_change_out.txt	Land use and parameter changes made by decision tables.

File	Content
simulation.out	Run log.
success.fin	Written when the run finishes normally.

See Checker and diagnostics for their contents.

Related files

- print.prt: the switches described on this page.
- file.cio: names the output folder.
- Output reference: overview of the output pages.

Sources

- print.prt reader: basin_print_codes_read.f90; switch type basin_print_codes in basin_module.f90
- Output folder: output_path_module.f90, read in readcio_read.f90
- Header writers called at startup: proc_open.f90, which calls output_landscape_init.f90, header_write.f90, header_sd_channel.f90 and the other header_*.f90 routines
- Files written every run: proc_bsn.f90, proc_hru.f90, main.f90.in
- Print window: time_control.f90

print.prt item	Fortran variable
nyskip, day_start, yrc_start, day_end, yrc_end, interval	pco%nyskip, pco%day_start, pco%yrc_start, pco%day_end, pco%yrc_end, pco%int_day
aa_int_cnt and end years	pco%aa_numint, pco%aa_yrs
csvout, use_obj_lbls, cdfout	pco%csvout, pco%use_obj_labels, pco%cdfout
crop_yld, mgtout, hydcon, fdcout	pco%crop_yld, pco%mgtout, pco%hydcon, pco%fdcout
object row <name>	pco%<group>%d, %m, %y, %a (for example hru_wb -> pco%wb_hru, channel_sd -> pco%sd_chan, basin_psc -> pco%recall_bsn)